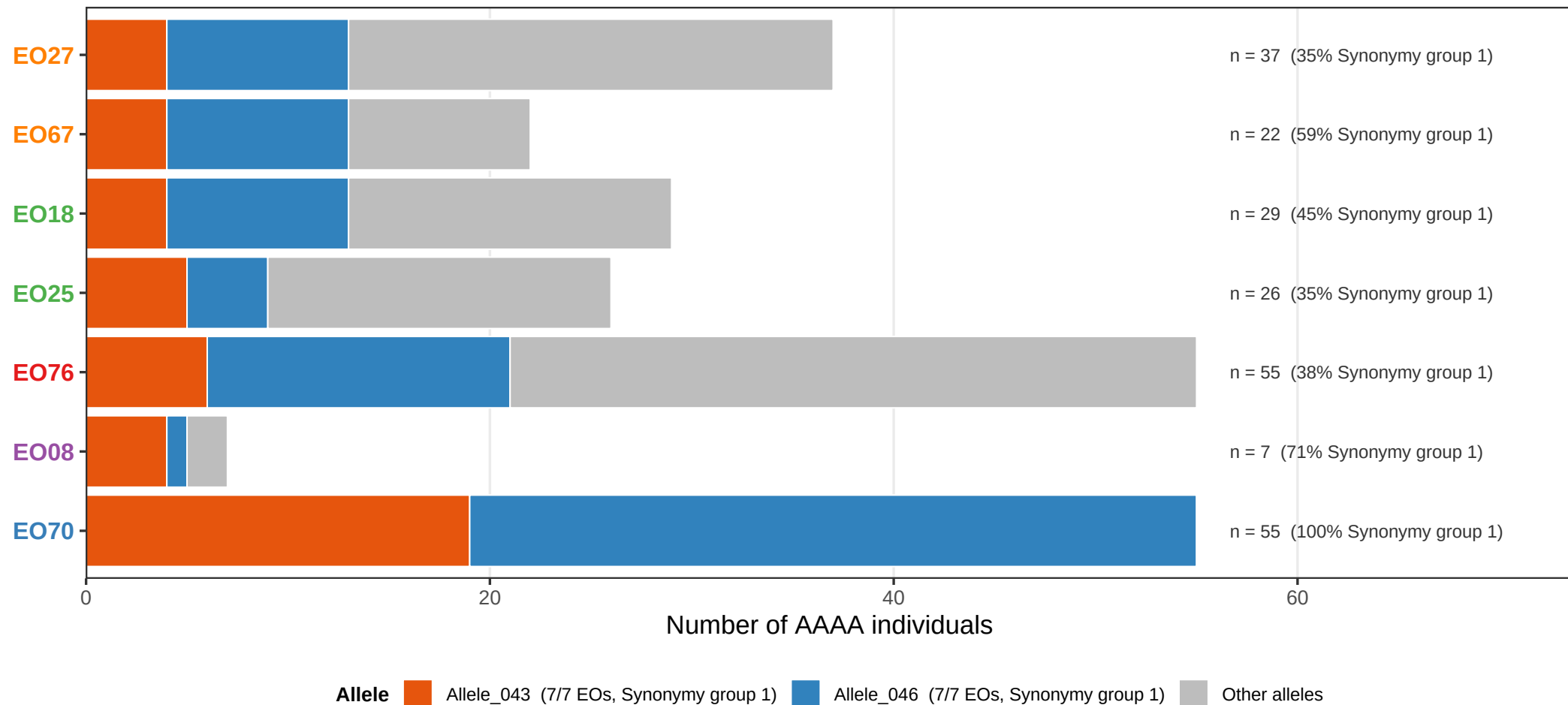


Allele identity of AAAA individuals per Element Occurrence

Each AAAA individual carries a single SRK allele across all four genome copies. Synonymy group 1 alleles (Allele_043, Allele_046) appear in AAAA individuals in $\geq 80\%$ of focus EOs (HV-identical – likely the same SI specificity hammered by drift across multiple EOs). EOs ordered by parent BL (total habitat area, largest at top) then by mean Drift_index within BL. Y-axis labels coloured by parent BL (Set1 palette).



Allele identity of AAAA individuals per Bottleneck Lineage

AAAA individuals aggregated across all BL-assigned EOs (n = 237 AAAA individuals).
Pan-BL Synonymy group 1 alleles (Allele_043, Allele_046) are present in AAAA individuals across every BL – confirms shared Synonymy group 1 fixation despite independent bottlenecks.
BLs ordered by total habitat area, then within-BL connectivity (largest at top). Y-axis labels coloured by BL (Set1 palette, matches LEPA_EO_spatial_clustering).

